

MAT499 / 599

Project 4: Image Classification

April 24, 2026

DUE: 2:30pm on Monday May 4th, 2026

Submission: Please submit this project to canvas as an iPython Notebook (.ipynb file). The project is worth 25% and 20% of the overall grade for MAT499 and MAT599, respectively. While the submission is a python script, also focus on presenting the questions logically (write them out!), explaining what the code in each block does using comments, and answering the word questions fully in text blocks within the notebook.

Task: This project will attempt to model the Wildfire Smoke dataset (presented here: <https://public.roboflow.com/object-detection/wildfire-smoke>). The data are images of wildfires taken from various surveillance towers and the targets are bounding boxes identifying the location of the smoke in the image. There are 516 training images, 147 validation images, and 74 test images. I have made the data available on a google drive here: <https://drive.google.com/drive/u/0/folders/1-iLL9m2NfXp0PaSnRrJ-NaqBX4p5Cfz>.

Prior to running the following set a seed to 499 or 599 (your enrolled class) so that your results are the same for every run of your notebook. You will need to set the seed for the numpy.random, random, and torch libraries.

Question 1

Load the data from the google drive and write a function that displays a training image chosen at random with its ground truth bounding box.

Question 2

Load a pretrained ResNet50 model from PyTorch. Train this model to predict bounding boxes on the wildfire images. You will have to change the output layer of the pretrained model to make this work. Use the training and validation loss to guide the number of epochs you need to find a model that performs well. Use a SmoothL1L loss function for your learning. State the training loss and validation loss of the final trained model.

Using the final model, calculate the mean intersection over union for the predicted bounding boxes for each of the training and validation images.

Question 3

Using the final model, predict the bounding boxes for the test dataset. Calculate the mean intersection over union for the test data.

Display the test image with the best intersection over union and include the predicted and the ground truth bounding boxes on the image.